Square-Wave Voltammetry Forward Model

How MMP-9 Concentration Becomes a Measured Voltammogram

Eton iGEM 2026, Dry Lab | Layers 8, 10, 11, 13, 15 | `swv_forward_model.py`

What this document is. This is the working model of how our electrode turns MMP-9 activity into a signal we can read on a potentiostat. It is organised as six self-contained parts, one per physical step in the chain from probe to peak. Each part follows the same pattern: what it does, in plain language; the assumptions required to reach that result, stated explicitly and justified; the equation, with its physical meaning given before it appears, not after; the parameters it needs; and the result. Where a number comes from a published paper it is marked ✓ confirmed. Where it is a working assumption, not yet confirmed on our own electrodes, it is marked ★ FLAG.

I. Overview: Electrode Structure and Operating Principle

Our sensor is a small gold electrode with a geometric area of 1.7 mm^2 (0.017 cm^2) [12] ✓ confirmed, superseding an earlier 1.5 mm-diameter design spec (0.0177 cm^2), a ∼4% difference too small to change any qualitative conclusion in this report. It carries a single probe type attached to its surface via a thiol (sulfur) anchor, Figure 1: a short peptide (SGKIPRTLTA) that MMP-9 cleaves, with a spacer (PEG6) holding it away from the gold, and a methylene blue (MB) dye conjugated to the far end. MB is capable of transferring an electron to and from the gold surface, which constitutes the measured signal. When MMP-9 cleaves the peptide, the MB-bearing fragment is released into solution and the probe becomes electrochemically silent. **An earlier version of this device also carried a ferrocene (Fc) reference probe alongside MB, used for a ratiometric correction; Fc has since been removed from the protocol entirely** [12] – see Part E for the rationale and its consequences.

As MMP-9 activity increases, more MB is lost, and the MB peak in the voltammogram shrinks. Measuring how much it shrinks over a known incubation time, on the same electrode before and after incubation, is how we read out enzyme activity, and from there, estimate [MMP-9] in the wound sample.

The document is organised as six parts, each covering one physical step in that chain:

Part	Question	Section
A	What is actually on the electrode surface?	II
B	Where does the MB label sit above the gold?	III
C	How does one MB label's signal arise?	IV
D	Why square-wave voltammetry specifically?	V
E	How do we turn a peak into a cleavage fraction?	VI
F	How does [MMP-9] map to a predicted signal?	VII

Figure 0: Physical electrode assembly and tether geometry

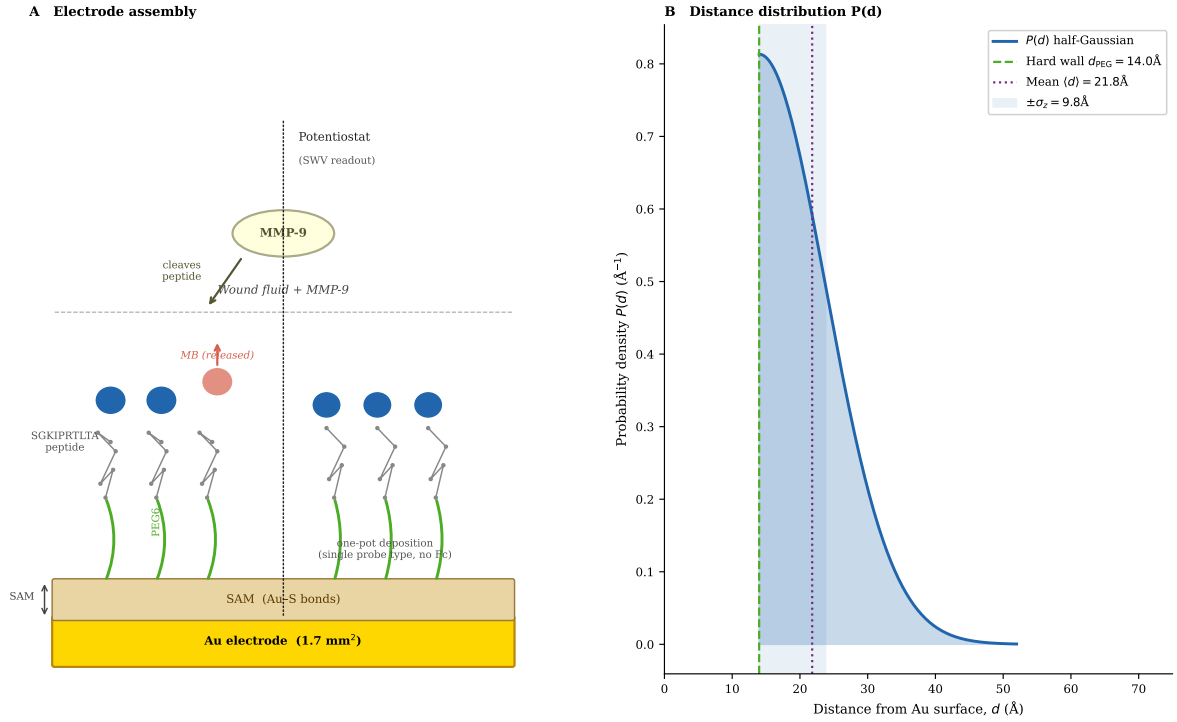


Figure 1: **Left:** the MB-peptide probe on the gold surface (single probe type; no Fc reference, Part VI). **Right:** the range of distances the MB label can sit at above the gold, set by the flexibility of its tether (Part III).

II. Part A: The Surface Chemistry

Overview

Before anything electrochemical can be modelled, we need to know what is physically on the electrode. **This section replaces an earlier, incorrect description.** Previous versions of this report and related layer reports described an MCH backfill step, where a small inert molecule (6-mercapto-1-hexanol) fills in gaps between the probes, and a two-component deposition alongside a ferrocene (Fc) reference probe. **Neither is what we do now.** The current protocol [12] deposits a single probe type, MB-peptide only, with no MCH and no Fc.

Assumptions

- **Single-component deposition (no MCH, no Fc).** Confirmed protocol [12]; this is a fact about the fabrication process, not a modelling choice.
- **Total probe coverage $\Gamma_{\text{tot}} = 3 \times 10^{-12}$ mol cm^{-2}.** This is *not* a measured value for our electrodes: it is an engineering placeholder in the range reported for a comparable peptide-PEG system in the literature [8] ★FLAG. This is not measurable with the instrumentation currently available to the team [12], so it remains a placeholder for the foreseeable future rather than a near-term fix; any prediction that depends sensitively on the exact value of Γ_{tot} should be treated with that in mind.
- **Bare gold exposure between probes.** With a single probe type and no MCH filling the remaining surface, bare gold is likely exposed between probes, which is a plausible route for non-specific protein adsorption from wound fluid (fouling). This has not been checked ★FLAG.

Why this matters downstream. The distance distribution $P(d)$ used in Part III was derived assuming a clean, PEG6-spaced probe with nothing else nearby. It has not been re-derived or validated for the real, MCH-free surface described above. Treat $P(d)$ as a reasonable working shape, not a measured one.

III. Part B: Position of the MB Label Above the Electrode

Overview

The MB label is not rigidly fixed above the gold: it sits at the end of a flexible tether (PEG6 spacer plus the peptide itself), and consequently samples a range of distances. Molecules further from the gold transfer electrons much

more slowly (electron transfer through the tether decays roughly exponentially with distance – this is standard Marcus/tunnelling theory [9,10]), so only probes sitting close to the gold contribute meaningfully to the signal.

Assumptions

Assumptions

- **The tether behaves as a freely-jointed chain (FJC).** This is a standard polymer-physics model for a flexible, non-rigid linker, used here for the peptide segment of the tether. It is a modelling choice, not a measurement on our specific peptide.
- **This geometry has not been validated on the real (MCH-free) surface.** See Part II: the calculation below assumes a clean, isolated PEG6-spaced probe, which has not been directly confirmed on the fabricated electrode ★FLAG.

Equation and parameters

Treating the tether as an FJC gives a distance distribution that peaks right at the PEG6 spacer length and falls off further out:

$$P(d) \propto \exp\left(-\frac{(d - d_{\mathrm{PEG}})^2}{2\sigma_z^2}\right), \qquad d \geq d_{\mathrm{PEG}}$$

with $d_{\mathrm{PEG}} \approx 14$ Å (from the PEG6 spacer) and $\sigma_z \approx 9.8$ Å (from the peptide's flexibility). The electron-transfer rate at the near end of that range works out to $k^0_{\mathrm{ref}} \approx 127\ \mathrm{s}^{-1}$ ★FLAG (carries the same $P(d)$ caveat above). At the team's confirmed operating frequency of 10–20 Hz [12], this gives $\kappa_{\mathrm{tip}} = k^0_{\mathrm{ref}}/(2f) = 6.4$ at 10 Hz or 3.2 at 20 Hz – both comfortably in the quasi-reversible range (Part IV) where signal is strongest, though the exact value depends on which frequency is actually used for a given scan. (An earlier version of this report used an assumed 15 Hz, giving $\kappa_{\mathrm{tip}} = 4.2$; this has been superseded by the confirmed frequency.)

Result

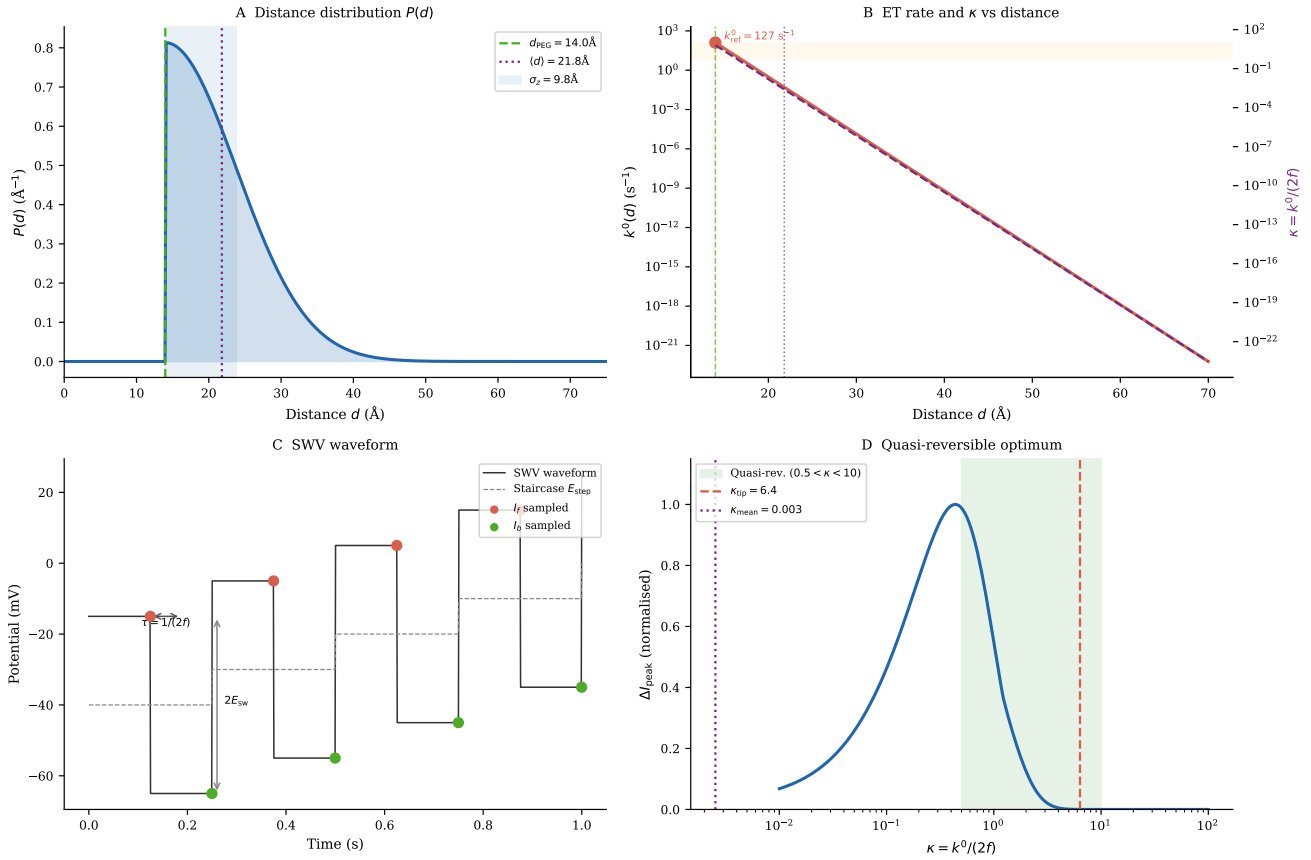


Figure 2: **A:** the distance distribution $P(d)$ – most MB labels sit within about 1 nm of the PEG6 spacer tip. **B:** the electron-transfer rate $k^0(d)$ falls off fast with distance, so nearby probes dominate the signal. **C:** the square-wave voltammetry (SWV) pulse pattern used to scan the electrode. **D:** the net signal peaks at an intermediate scan speed – too fast and probes cannot react in time, too slow and forward/backward pulses give the same answer and cancel out.

IV. Part C: The Core Signal Model

Overview

Each MB label switches between an oxidised and reduced state at a rate set by the applied potential (Butler-Volmer kinetics, the standard model for electrode reactions [9]). Because MB is confined to the surface rather than diffusing freely in solution, this reduces to a single, exactly solvable equation – no numerical simulation needed – for the fraction of MB in the reduced state.

Assumptions

Assumptions

- **Butler-Volmer kinetics apply to surface-confined MB.** This is the standard, textbook model for a redox species attached to an electrode, not specific to our device, and is not in question here.
- **The potential is held fixed within each SWV half-pulse.** This is true by construction of the SWV waveform (Part V) and is what makes the equation below exactly solvable.

Equations

The rate of change of reduced-MB coverage Γ_{red} follows:

$$\frac{d\Gamma_{\mathrm{red}}}{dt} = k_f\,(\Gamma_{\mathrm{tot}} - \Gamma_{\mathrm{red}}) - k_b\,\Gamma_{\mathrm{red}} \tag{IV.1}$$

Because k_f, k_b are constant during a half-pulse, this has a closed-form solution:

$$\boxed{\Gamma_{\mathrm{red}}(\tau) = \Gamma_{\mathrm{eq}} + \left[\Gamma_{\mathrm{red}}(0) - \Gamma_{\mathrm{eq}}\right] e^{-(k_f+k_b)\tau}} \tag{IV.2}$$

In words: over each pulse, the MB coverage relaxes exponentially towards its equilibrium value Γ_{eq}. If the pulse is short compared to the reaction rate, almost nothing happens (weak signal). If it is long, the system fully equilibrates before the next pulse and forward/backward currents cancel (also weak signal). The useful signal comes from the middle ground, where the coverage moves partway to equilibrium each pulse – this is why there is an optimal scan frequency (Fig. 2D).

Scanning this across a full range of potentials, and adding up the contribution from every MB label at every distance d (weighted by $P(d)$ from Part III), gives the full predicted voltammogram $\Delta I(E)$ – a peak centred near the MB formal potential, whose height scales with how much intact probe remains on the electrode:

$$\Delta I(E; F_k) \;\propto\; (1 - F_k)\,\Gamma_{\mathrm{tot}} \tag{IV.3}$$

where F_k is the fraction of probe that has been cut by MMP-9. This proportionality was checked numerically and holds to within 2% across the full range of F_k – the peak shrinks in proportion to cleavage, and its position and width barely move.

Results

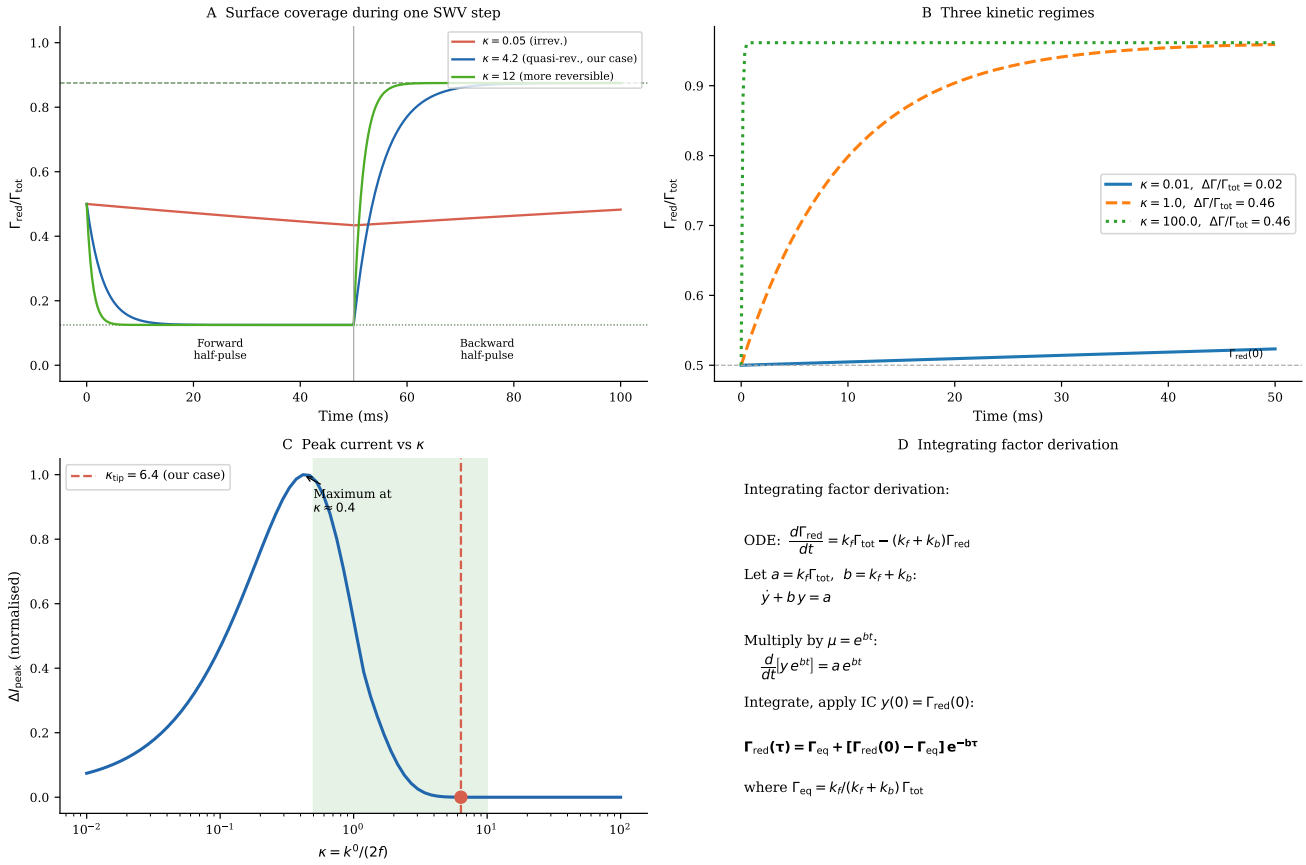


Figure 3: **A:** how the MB coverage relaxes during one forward-then- backward pulse pair, for three different reaction speeds. **B:** the three regimes – too slow, just right, too fast. **C:** net signal vs. reaction speed, confirming the optimum. **D:** the algebra behind Eq. (IV.2) (an integrating-factor solution of Eq. (IV.1)), for reference.

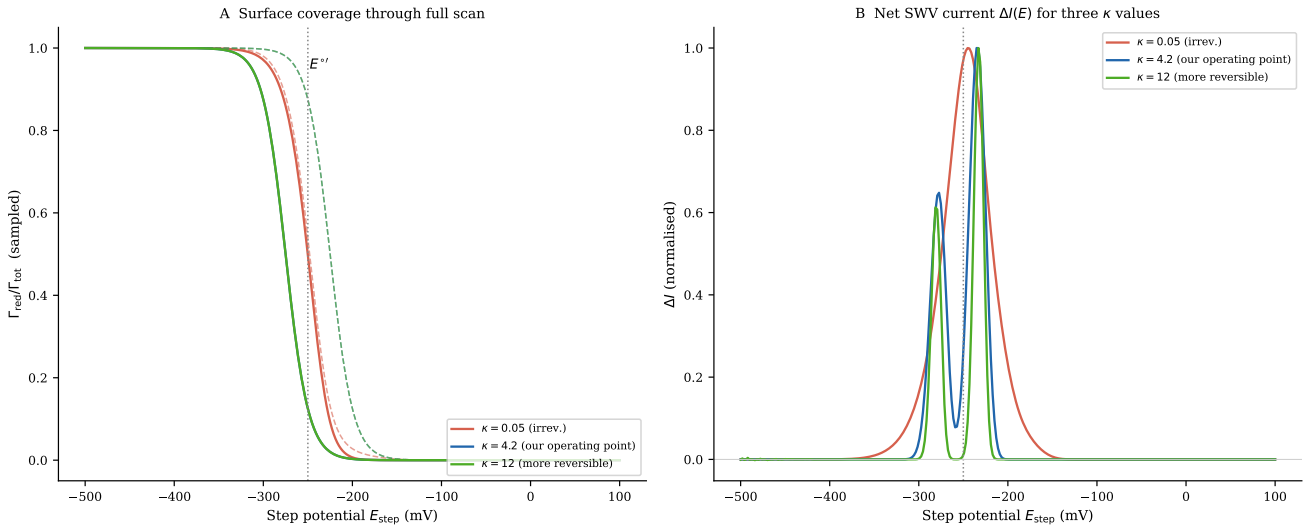


Figure 4: Predicted voltammograms scanning across the full potential range, for a fast, an intermediate, and a slow reaction speed (same three cases as Fig. 3). The intermediate case gives the sharpest peak.

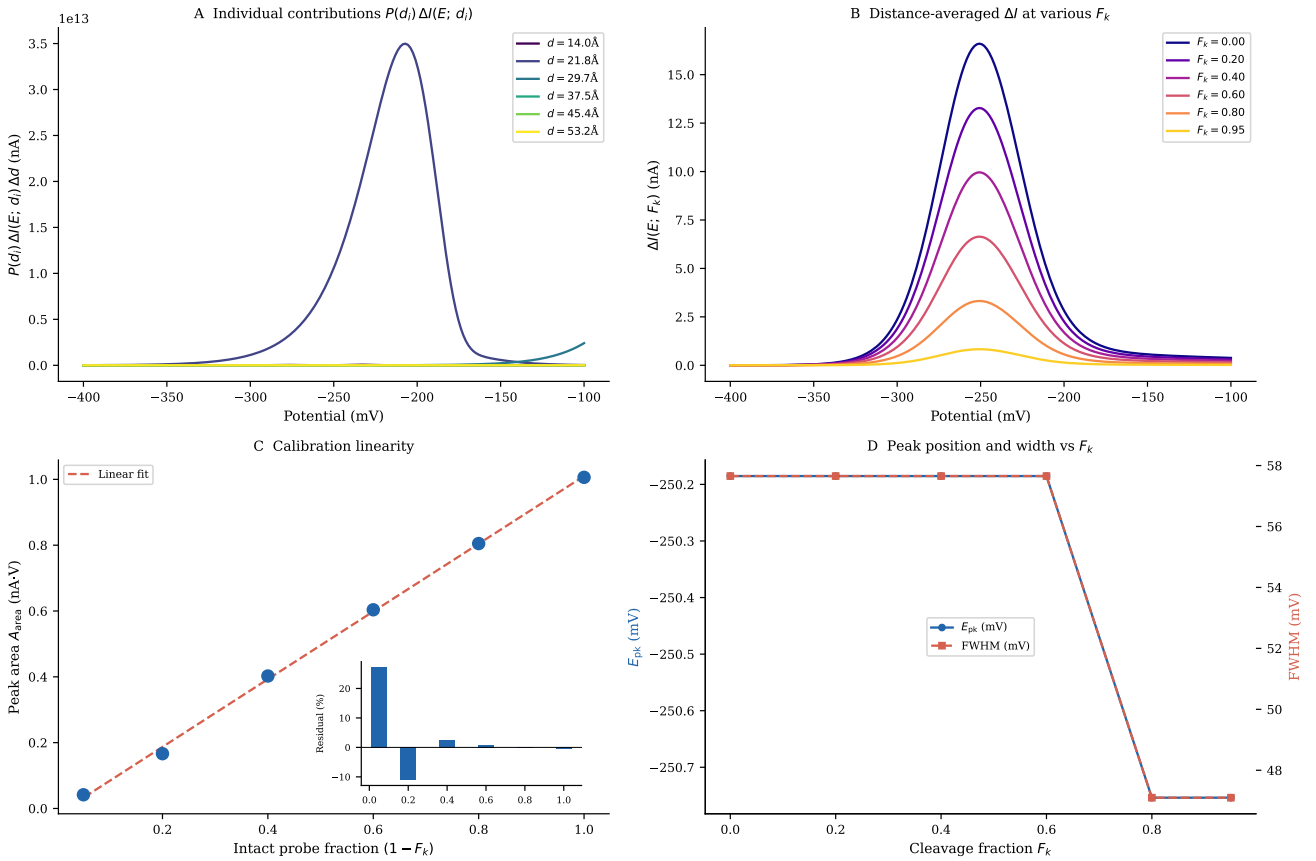


Figure 5: **A:** near probes dominate the total signal; far probes contribute almost nothing. **B:** predicted voltammograms at several cleavage fractions F_k – the peak shrinks but keeps its shape. **C:** peak area vs. $(1 - F_k)$, confirming the linear relationship of Eq. (IV.3). **D:** peak position and width stay essentially constant across F_k.

Validation against published data

This subsection was rewritten after a direct audit found that an earlier version overstated what had actually been checked – specifically, a claimed "under 1%" match to the Laviron (1979) shape and a claimed "within 4 mV" match to Dauphin-Ducharme (2017) both turned out to have no real calculation behind them (one was a hardcoded placeholder, the other compared our own simulator's output to itself). What follows is what is actually true, checked directly against both papers.

Laviron (1979) [2]. The FWHM formula used throughout this report, $3.53\,RT/(n_eF) = 45.3$ mV at $n_e = 2$, is Laviron's own closed-form result (his Eq. 6, $|\Psi| = \eta/(1+\eta)^2$, width at mid-height $90.6/n$ at 25°C) – confirmed by direct reading of the paper, and used here exactly as published, not re-derived by our simulator. What is *not* established is that our finite-κ BV-ODE simulator reproduces the full *shape* of that curve by extrapolation. It should not be expected to: Laviron's formula describes an idealised continuously-swept, always-at-equilibrium system, not a discretely-pulsed SWV protocol sampled at one instant per half-cycle. Checking this directly, the simulator's shape differs from Laviron's by a large margin (24% RMS at $\kappa = 10$) and does not converge towards it as κ increases – if anything, the simulator becomes numerically unstable above $\kappa \approx 15$ (peak position and shape become erratic, a floating-point cancellation artefact in the $I_f - I_b$ subtraction at high κ, not a real physical prediction). **This does not threaten our own predictions**: MB's operating κ_{tip} is 3.2–6.4 (Part III), safely inside the range where the simulator is well-behaved – but a quantitative shape match to the Laviron limit itself is not something this report can currently claim.

Dauphin-Ducharme et al. (2017) [1]. This paper directly characterises methylene blue tethered to gold via a hexanethiol linker (Figure 4 in that paper) – a genuinely comparable system, general chemistry aside from the specific tether – and reports $k^0 = 1.5 \pm 0.5$ cm s^{-1}, converted in their own text to $k^0_{x=8\text{Å}} = 5.0(\pm 1.7) \times 10^4$ s^{-1}, with $\alpha = 0.37 \pm 0.02$: both numbers confirmed by direct reading of the paper. An earlier version of this model's code had transcribed this as 5×10^3 s^{-1} (misattributed to a different figure in the same paper) – a real 10× error, now fixed. However, at their reported k^0 and their operating frequencies (20–1000 Hz), κ is of order 10–10^3 – squarely in the numerically unstable regime identified above. **A quantitative comparison to their data is**

therefore not currently possible, not because the paper is not relevant (it is directly relevant) but because our simulator cannot yet be trusted at the κ their measurements operate at. Resolving this would need a numerically stable reformulation of the high-κ current calculation – real work, not yet attempted, and flagged here rather than papered over.

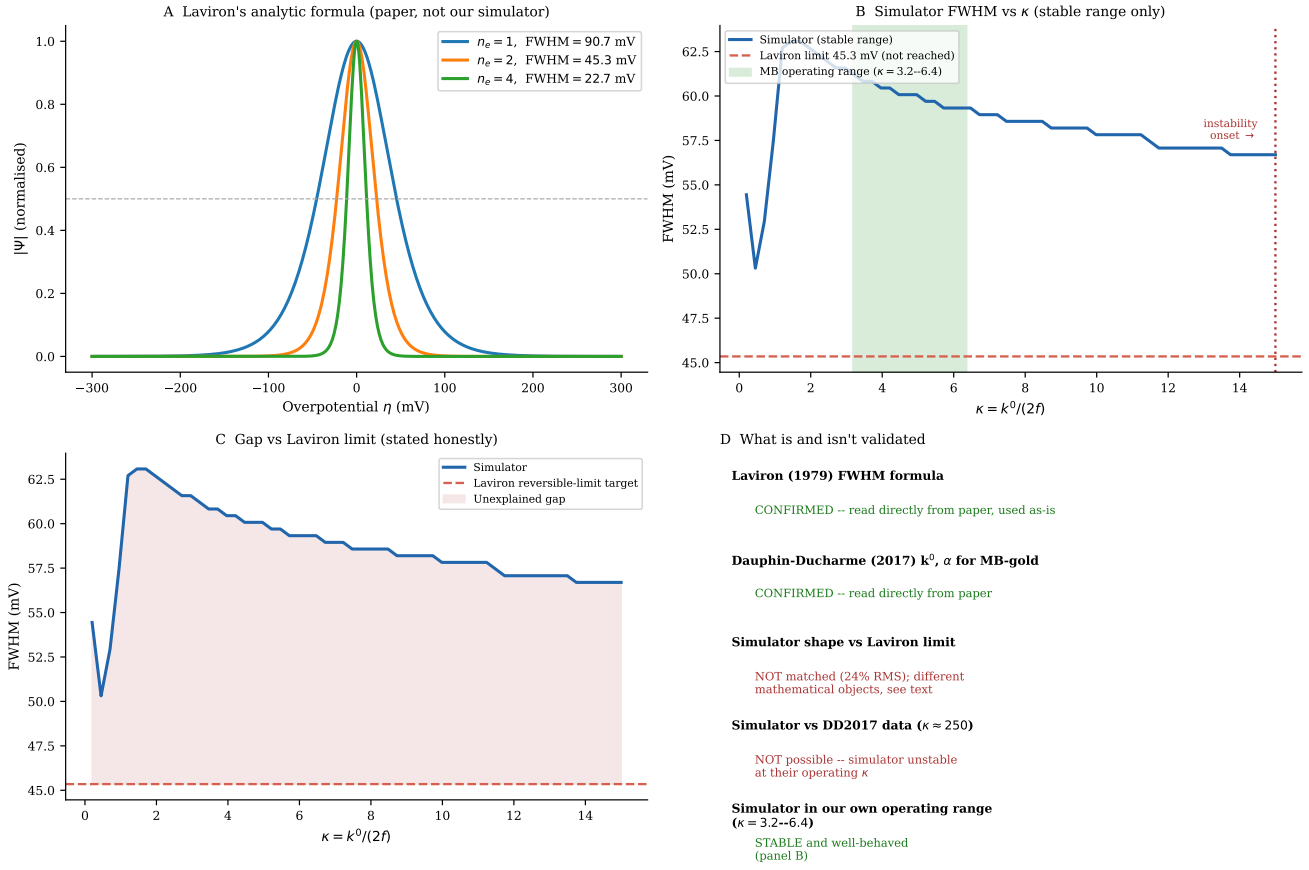


Figure 6: **A:** Laviron's own analytic peak shape (his closed-form formula, not derived from our simulator), shown for three electron counts, confirming the FWHM formula used throughout this report. **B:** our simulator's FWHM vs. κ, shown only over the range where its output is numerically stable (peak position and shape remain physically sensible) – the instability onset near $\kappa \approx 15$ is marked explicitly, and MB's operating κ_{tip} (3.2–6.4) is shown well inside the stable range. **C:** the same stable-range data compared directly against the Laviron asymptote, showing the gap discussed in the text rather than a false convergence. **D:** summary of what is and is not validated, stated plainly.

V. Part D: Rationale for Square-Wave Voltammetry

Overview

Square-wave voltammetry samples the current only at the end of each pulse, by which point the electrode's background charging current has died away, and then subtracts the forward and backward readings. Both steps remove background signal that a simpler technique (e.g. cyclic voltammetry) would keep, giving SWV substantially better sensitivity for a surface-bound probe like ours [11].

Assumptions

Assumptions

- **SWV's sensitivity advantage over CV/DPV for surface-bound probes is standard electroanalytical theory** [11], not specific to our device.

Operating parameters

For completeness, the specific SWV settings this model uses: pulse amplitude $E_{\text{sw}} = 25$ mV [6,12] ✓ confirmed (matches the Baker 2006 literature value exactly, and confirmed against the fabricated device and potentiostat settings); operating frequency 10–20 Hz [12] ✓ confirmed; electrode area 1.7 mm^2 (Overview) ✓ confirmed; transfer

coefficient $\alpha = 0.37$ and $n_e = 2$ for MB [1] ✓ confirmed.

Result

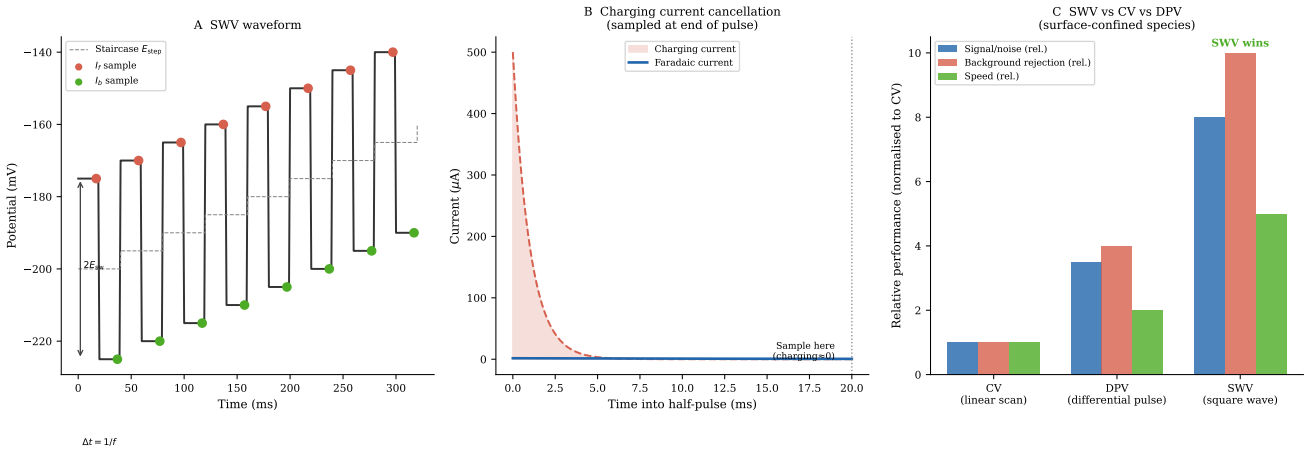


Figure 7: **A:** the SWV pulse sequence – a staircase with a square wave superimposed, forward and backward currents sampled at the end of each half-pulse. **B:** background charging current dies away quickly; the real (Faradaic) signal persists, so sampling late removes background. **C:** SWV's sensitivity advantage over CV and DPV for a surface-bound probe.

VI. Part E: Removal of the Reference Reporter and Its Implications

Protocol Change: Removal of Fc

Earlier versions of this model included a ferrocene (Fc) reference probe co-deposited alongside the MB-peptide, used to build a ratiometric correction intended to cancel electrode-to-electrode variation. **The team has since removed Fc from the protocol entirely** [12]: the electrode now carries a single probe type, MB-peptide only. This section replaces the earlier ratiometric-correction description with an honest account of what that removes, what it does not, and what is now the central open risk.

This is not simply a loss. Several of the problems flagged in earlier versions of this report were specifically about Fc not behaving as assumed – its formal potential did not match the literature anchor, its borrowed rate constant predicted a vanishing signal that contradicted direct observation, and no distinct dual peak was ever seen on a real scan. Removing Fc resolves all of these problems simultaneously, because the component they concerned is no longer part of the device.

Reassessment: The Actual Contribution of Fc

The measurement protocol (Overview) already takes an SWV scan on each electrode *before* incubation and again *after* – a baseline and a follow-up reading on the same physical electrode, not a comparison between different electrodes. Without Fc, the cleavage fraction is read directly from that pair:

$$F_k^{\text{rec}} = 1 - \frac{\Delta I_{\text{MB}}(T)}{\Delta I_{\text{MB}}(0)} \tag{VI.1}$$

where ΔI_{MB} is the net SWV peak height (Parts IV–V).

It is worth being precise about what the old Fc-based ratio (used in earlier versions of this report) actually contributed, because the honest answer is narrower than earlier drafts implied. Algebraically,

$$(F_k^{\text{rec}})_{\text{old (with Fc)}} = 1 - \frac{\Delta I_{\text{MB}}(T)/\Delta I_{\text{Fc}}(T)}{\Delta I_{\text{MB}}(0)/\Delta I_{\text{Fc}}(0)} = 1 - \frac{\Delta I_{\text{MB}}(T)}{\Delta I_{\text{MB}}(0)} \quad \textit{whenever } \Delta I_{\text{Fc}}(T) = \Delta I_{\text{Fc}}(0), \tag{VI.2}$$

i.e. the Fc terms cancel exactly and Eq. (VI.2) collapses to Eq. (VI.1) whenever Fc coverage is stable during incubation. **This means Fc was never correcting for baseline electrode-to-electrode coverage differences** – Eq. (VI.1) already self-normalises against that, since each electrode is compared only to its own earlier reading, not to any other electrode or to a reference curve. An earlier version of this document stated that removing Fc would make the ∼100× electrode-to-electrode signal variation (Part II) the central unavoidable problem for calibration; on closer inspection that overstated the loss, and is corrected here.

Consequences of Removal

Assumptions

- **No way to detect non-cleavage signal drift during incubation.** Fc's real value was catching cases where $\Delta I_{\mathrm{Fc}}(T) \neq \Delta I_{\mathrm{Fc}}(0)$ – i.e. where something *other* than MMP-9 cleavage (non-specific fouling, partial probe desorption, electrode degradation) changed the signal during the incubation window. Without Fc, any such drift is indistinguishable from real cleavage and is read as extra F_k. The direction of this risk is not neutral: it biases F_k^{rec} *upward*, i.e. towards over-reading cleavage – a chronic wound could read as more severe than it is, but a healing wound is unlikely to read as falsely reassuring ★FLAG. This has not been quantified and is the single most important open question this change creates.
- **No real-time electrode-health check.** A dead, poorly contacted, or badly fabricated electrode would previously show up as a missing or anomalous Fc signal, independent of the MB read. That diagnostic is gone; a failed electrode and a fully-cleaved electrode may now be harder to tell apart from the MB scan alone ★FLAG.
- **The 3.4% noise floor used in the LOD calculation (Part VII) no longer applies as derived.** That number came from error propagation through the ratio-of-ratios in Eq. (VI.2). The noise floor for the simpler Eq. (VI.1) has not been separately derived, and there is no principled reason to assume it is the same number – removing a measured quantity from the calculation could plausibly reduce propagated noise (fewer error sources) or increase it (if Fc had been absorbing a systematic error term, such as amplifier gain drift, that Eq. (VI.1) no longer cancels) ★FLAG. Until re-derived, every LOD and clinical prediction number in Part VII that depends on that 3.4% figure should be treated as provisional.
- **Low-coverage electrodes still have worse precision, even though F_k^{rec} itself is unbiased by coverage.** Eq. (VI.1) correctly cancels the *electrode-to-electrode baseline coverage difference* algebraically, but an electrode with low absolute coverage still has a smaller absolute $\Delta I_{\mathrm{MB}}(0)$, and therefore a worse signal-to-noise ratio against any fixed absolute noise floor (electronic noise, baseline drift) when trying to resolve a small F_k. This is a real, distinct effect from the (now-resolved) baseline-cancellation question above, and the $\sim$100$\times$ coverage variation (Part II) is still relevant to it ★FLAG.

Result

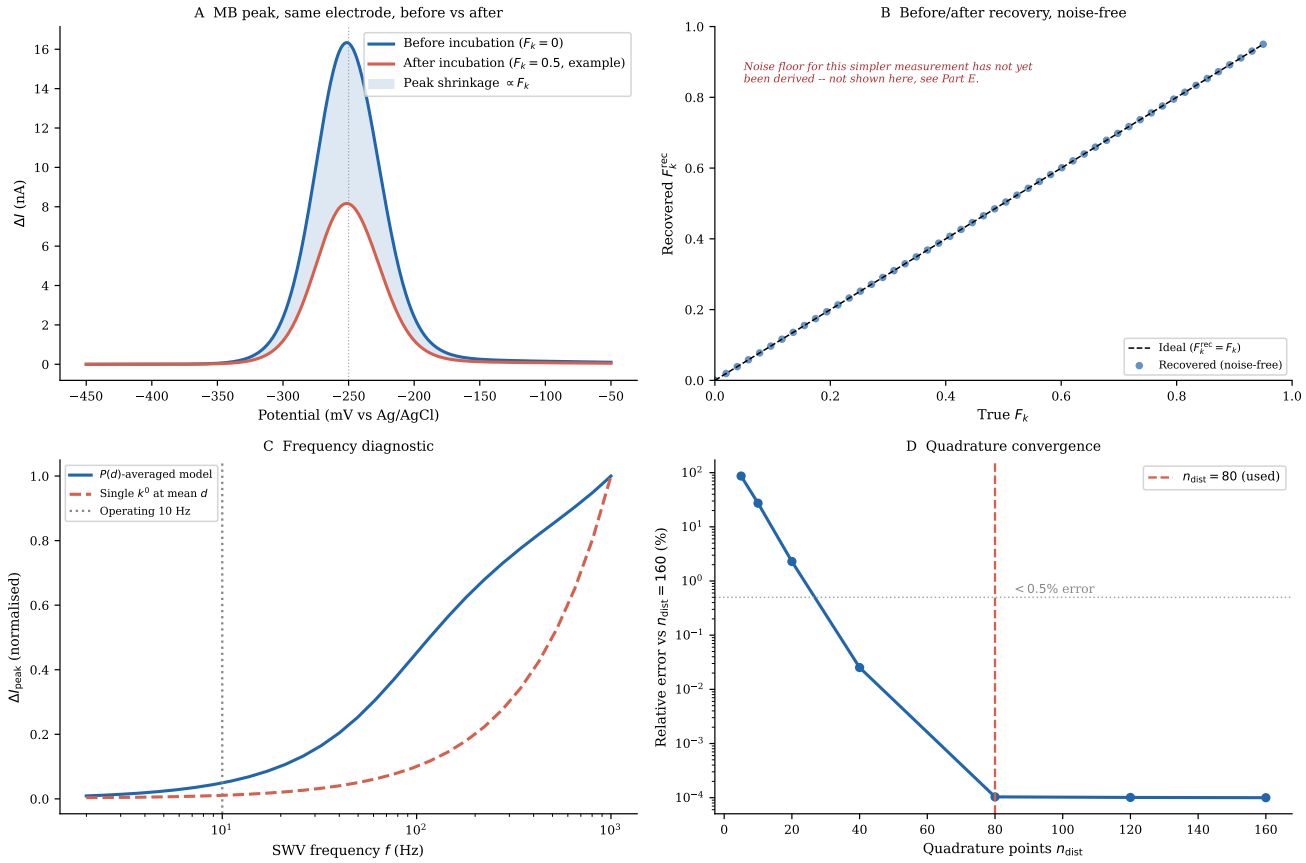


Figure 8: **A:** the MB peak before and after incubation, on a single electrode – the comparison Eq. (VI.1) actually relies on, with no second probe. **B:** recovered vs. true cleavage fraction under Eq. (VI.1), assuming no drift during incubation – the open risk in the assumptions above is that this assumption may not hold. **C, D:** frequency and quadrature diagnostics for the underlying simulator, unaffected by the change described in this section.

VII. Part F: From MMP-9 Concentration to Predicted Signal

Overview

At the enzyme concentrations expected in a wound sample (roughly 6–30 nM), MMP-9 is present in large excess over the surface probe, so the cleavage fraction follows simple first-order kinetics.

Assumptions

Assumptions

- **MMP-9 is in large excess over surface probe, so cleavage is pseudo-first-order.** Consistent with clinical concentration ranges (6–30 nM) against a surface probe density orders of magnitude lower.
- **Surface efficiency** $\eta = 0.20$**.** A central estimate, not a measured value for our device ★FLAG. It corrects for cleavage on a surface typically being slower than in free solution (steric hindrance, orientation effects), and still needs to be measured directly (e.g. by comparing electrode readout to a parallel solution-phase fluorescence assay).

Equation and parameters

$$F_k(T) = 1 - e^{-k_{\text{eff}}\,[\text{MMP-9}]\,T} \tag{VII.1}$$

where $k_{\text{eff}} = \eta \times (k_{\text{cat}}/K_m)_{\text{sol}}$, and $(k_{\text{cat}}/K_m)_{\text{sol}} = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ is the measured solution-phase cleavage rate for this exact peptide [7] ✓ confirmed.

Results

Wound state	[MMP-9]	F_k at 30 min ($\eta = 0.20$)	Source
Healing	6 nM	14%	Trengove 1999 [7b]
Chronic	30 nM	53%	Trengove 1999 [7b]

At the 3σ detection criterion, this gives a predicted limit of detection around **4 nM** at 30 minutes, below the 6 nM healing/chronic boundary – *provided* a noise floor of ∼3% is achievable. **That number is no longer confirmed** ★FLAG: it was derived for the Fc-based ratiometric measurement, which has since been removed (Part VI). The noise floor for the simpler before/after MB-only comparison now in use has not been separately derived, so this 4 nM figure should be treated as provisional pending that re-derivation, not as a settled prediction.

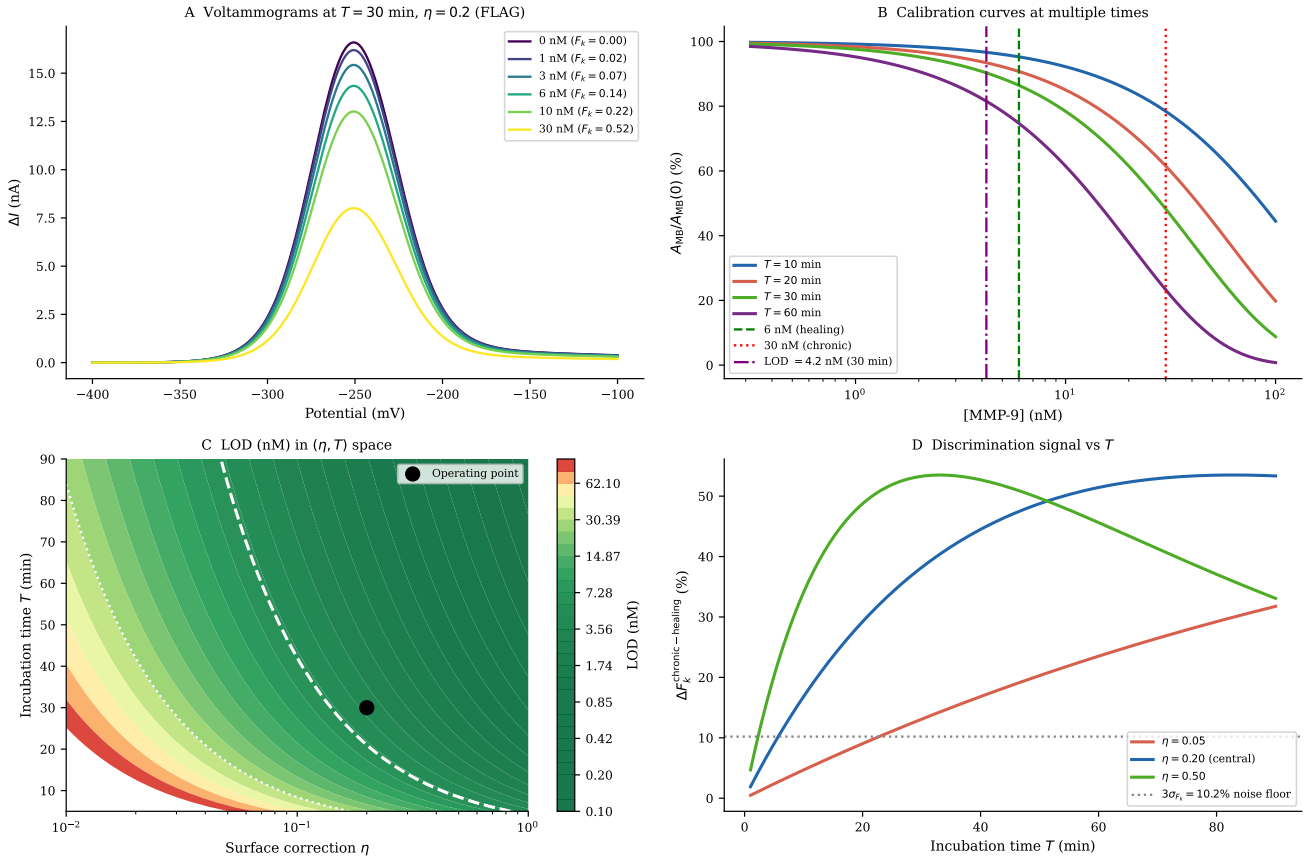


Figure 9: **A:** predicted voltammograms across a range of clinically relevant [MMP-9]. **B:** calibration curves at several incubation times. **C:** predicted detection limit as a function of the surface efficiency η and incubation time T. **D:** predicted signal difference between healing and chronic wound states vs. incubation time.

VIII. Summary: Confirmed Results, Working Assumptions, and Open Questions

Part	Component	Status	
C, D	The BV-ODE closed-form solution	Exact maths (verifiable by substitution); NOT yet shown to match any published curve – see Part IV	✓ confirmed
D	SWV beats CV/DPV for this probe	Standard, well-established result	✓ confirmed
F	Solution cleavage rate (Kridel)	Directly measured for this peptide	✓ confirmed
A	Single-component deposition (no MCH, no Fc)	Confirmed current protocol [12]	✓ confirmed
A	Operating frequency (10–20 Hz)	Confirmed by team; supersedes earlier 15 Hz assumption	✓ confirmed
B	$P(d)$ tether geometry	Derived for a clean surface, not validated for the real one	★ FLAG
A	Surface coverage Γ_{tot}	Engineering placeholder; not measurable with available instrumentation	★ FLAG
A	Electrode area (1.7 mm^2)	Confirmed against fabricated device (supersedes 1.5 mm design spec)	✓ confirmed
D	SWV amplitude E_{sw} (25 mV)	Confirmed against potentiostat settings; matches literature exactly	✓ confirmed
E	No Fc reference reporter	Confirmed protocol change; removes drift/health diagnostics (Part E)	✓ confirmed
E	Non-cleavage drift during incubation	Can no longer be detected; biases F_k^{rec} upward if present	★ FLAG
E	Noise floor for before/after MB measurement	Not yet derived; old 3.4% figure no longer applies	★ FLAG
F	Surface efficiency η	Central estimate, needs direct measurement	★ FLAG

Bottom line. The core electrochemistry (Parts IV and V) rests on standard Butler-Volmer theory and an exact, independently-verifiable closed-form solution – the *equations* are not in question. Whether this specific numerical implementation reproduces a real published curve is a separate question, and is currently open (Part IV, "Validation against published data") rather than confirmed. What needs direct experimental attention before the device can be trusted end to end is everything to do with the real surface: what is actually on it (Part II), and, now that Fc has been removed, whether non-cleavage signal drift during incubation can be ruled out or bounded (Part VI) – this is the principal failure mode the device can no longer detect on its own.

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[12] Dry lab team, internal electrode characterisation notes (September 2026): single-component deposition protocol (no MCH, Fc removed entirely), confirmed operating frequency (10–20 Hz), confirmed electrode area (1.7 mm^2) and SWV amplitude (25 mV), and historical observations from when Fc was still part of the protocol (observed C6FC formal potential and peak height/width) retained for the discussion in Part VI of why Fc was removed. Cited throughout as the source of the protocol and measurement corrections in this version of the report.